

FURTHER MECHANICS / CENTRE OF MASS

Centre of Mass of Polygonal Laminae

Questions by topic

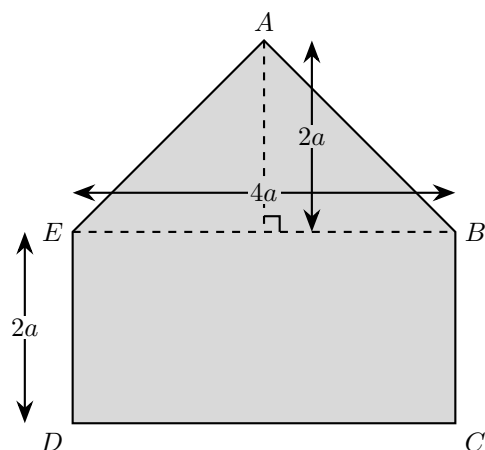
10 questions · 91 marks

Instructions

Unless a question states otherwise, take $g = 9.8 \text{ m s}^{-2}$ wherever a numerical value of g is required. Solutions also give final answers for $g = 10 \text{ m s}^{-2}$ where these differ.

Questions

Question 1	7 marks	2
Question 2	9 marks	3
Question 3	8 marks	4
Question 4	11 marks	5
Question 5	10 marks	6
Question 6	8 marks	7
Question 7	7 marks	8
Question 8	8 marks	9
Question 9	11 marks	10
Question 10	12 marks	11



1. A uniform plane lamina is in the shape of the pentagon $ABCDE$, as shown shaded in the diagram above.

- $BCDE$ is a rectangle
- $BE = 4a$ and $DE = 2a$
- triangle ABE is isosceles with perpendicular height $2a$

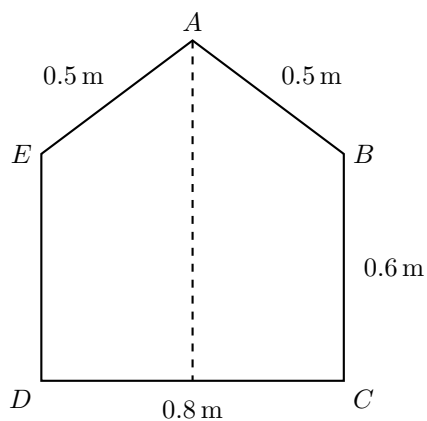
(a) Show that the distance of the centre of mass of the lamina from CD is $\frac{14a}{9}$. [5]

The mass of the lamina is M .

The lamina is suspended by two light vertical strings, one attached to the lamina at D and the other attached to the lamina at E .

The lamina hangs freely in equilibrium, with DE horizontal.

(b) Using part (a), find, in terms of M and g , the tension in the string attached at E . [2]



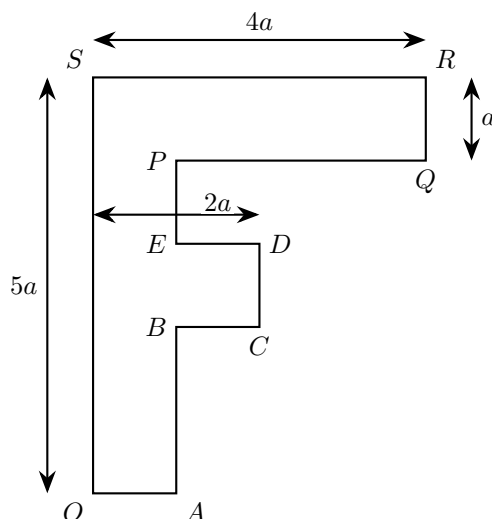
2. $ABCDE$ is a uniform lamina formed by joining an isosceles triangle ABE to the top of a rectangle $BCDE$, where $AB = AE = 0.5$ m, $BC = DE = 0.6$ m and $CD = BE = 0.8$ m, as shown in the diagram. The centre of mass of the lamina is G .

(a) Explain why G lies on the dashed line. [1]

(b) Show that the perpendicular distance of G from CD is 0.38 m. [4]

The lamina is freely suspended from the point B .

(c) Determine the acute angle that CD makes with the horizontal, stating which of C or D is higher. [4]



3. A letter F from a shop sign is modelled as a uniform plane lamina in the shape shown in the diagram above.

The left-hand vertical strip has width a and height $5a$. The top horizontal strip has length $4a$ and thickness a . The middle horizontal strip has length $2a$ and thickness a , and $AB = 2a$.

Using the model,

- (a) find, in terms of a , the distance of the centre of mass of the letter F , from

(i) OS

(ii) OA

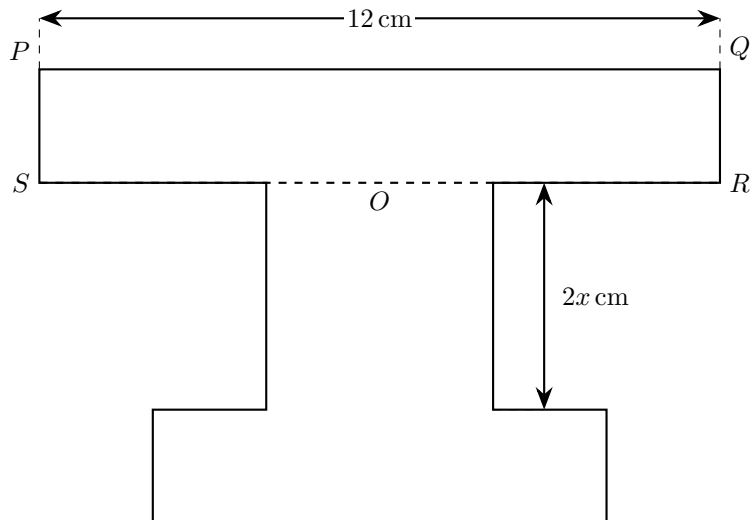
[6]

The letter F is freely suspended from O and hangs in equilibrium. The angle between OS and the downward vertical is α .

Using the model,

- (b) find the exact value of $\tan \alpha$.

[2]



4. A template T consists of a uniform plane lamina made from three rectangles, as shown in the diagram. The upper rectangle $PQRS$ has $PQ = 12$ cm and $PS = 2$ cm. The point O is the midpoint of SR . A rectangle of width 4 cm and height $2x$ cm is attached centrally below SR , and a rectangle of width 8 cm and height 2 cm is attached centrally below this rectangle. The whole template is symmetrical about the line through O perpendicular to SR , where $x > 1$.

(a) Show that the centre of mass of T is a distance

$$\frac{x^2 + 4x - 1}{x + 5} \text{ cm}$$

below SR .

[7]

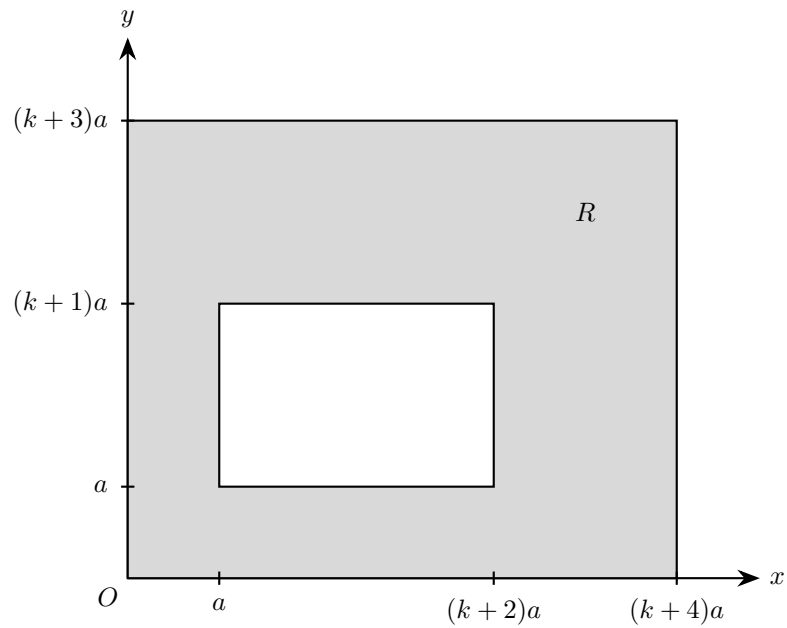
The template T is freely suspended from the point P and hangs in equilibrium.

Given that $x = 2$ and that θ is the angle that PQ makes with the horizontal,

(b) show that

$$\tan \theta = \frac{42}{25}$$

[4]

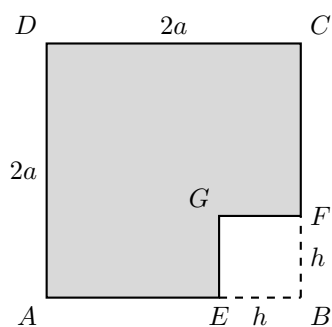


5. A uniform lamina R is bounded externally by the rectangle with vertices $(0, 0)$, $((k+4)a, 0)$, $((k+4)a, (k+3)a)$ and $(0, (k+3)a)$, where $k > 0$.

A rectangular hole with vertices (a, a) , $((k+2)a, a)$, $((k+2)a, (k+1)a)$ and $(a, (k+1)a)$ is removed, as shown in the diagram.

Determine the range of values of k for which the centre of mass of R lies in the hole.

[10]

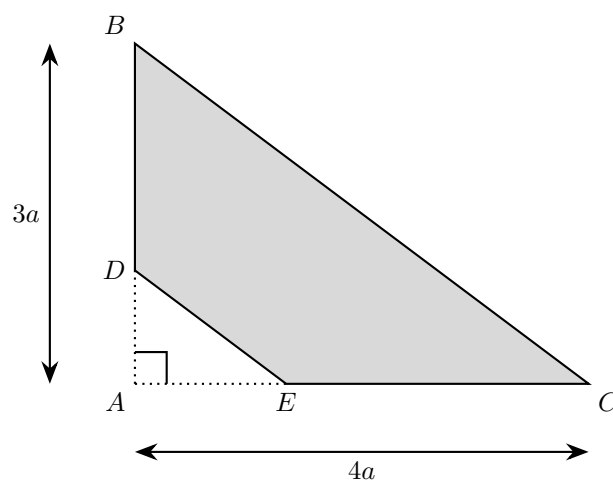


6. A uniform lamina is formed by removing the square $EBFG$ from a square lamina $ABCD$, where $AB = BC = CD = DA = 2a$ and $EB = BF = h$, with $0 < h < 2a$ (see diagram).

- (a) Find the distance of the centre of mass of the remaining lamina from AD and from AB , giving your answers in terms of a and h . [5]

The lamina is placed vertically on its edge AE on a horizontal plane.

- (b) Find, in terms of a , the set of values of h for which the lamina remains in equilibrium. [3]

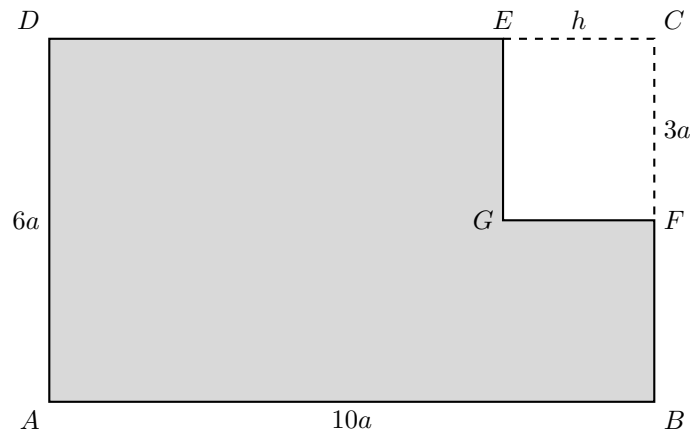


7. A uniform lamina is formed from the right-angled triangle ABC by removing the triangle ADE , as shown in the diagram. The triangle ABC has $AB = 3a$, $AC = 4a$ and $\angle BAC = 90^\circ$. The point D lies on AB and satisfies $AD : AB = 1 : k$. The point E lies on AC , and DE is parallel to BC .

- (a) Taking A as the origin, with AC as the x -axis and AB as the y -axis, find the coordinates of the centre of mass of the remaining lamina $DBCE$ in terms of a and k . [4]

When the lamina $DBCE$ is freely suspended from C , the edge CE makes an angle θ with the downward vertical, where $\tan \theta = \frac{21}{52}$.

- (b) Find the value of k . [3]



8. $ABCD$ is a uniform rectangular lamina with $AB = 10a$ and $BC = 6a$. Point E lies on DC with $CE = h$, where $0 < h < 10a$, and point F lies on BC with $CF = 3a$. The rectangle $CEGF$ is removed, where G is the fourth vertex of the rectangle (see diagram).

(a) Show that the distance of the centre of mass of the remaining lamina from AD is

$$\frac{h^2 - 20ah + 200a^2}{2(20a - h)}$$

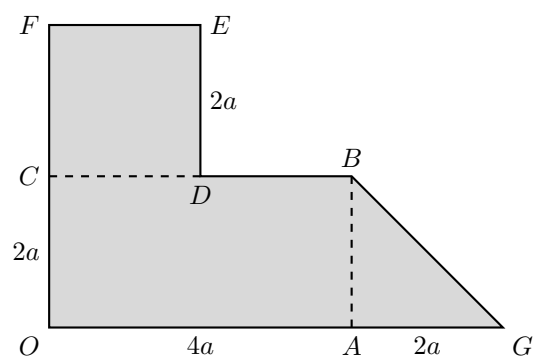
and find a corresponding expression for the distance of the centre of mass from AB .

[5]

When the lamina is suspended from the point D , the edge DA makes an angle θ with the downward vertical, where $\tan \theta = \frac{13}{12}$.

(b) Find, in terms of a , the two possible values of h .

[3]



9. The uniform plane lamina shown in the diagram is formed from the rectangle $OABC$, the square $CDEF$ and the triangle ABG . The rectangle has dimensions $4a$ by $2a$, the square has side $2a$ and $AG = 2a$. Take O as the origin, with OA as the positive x -axis and OC as the positive y -axis.

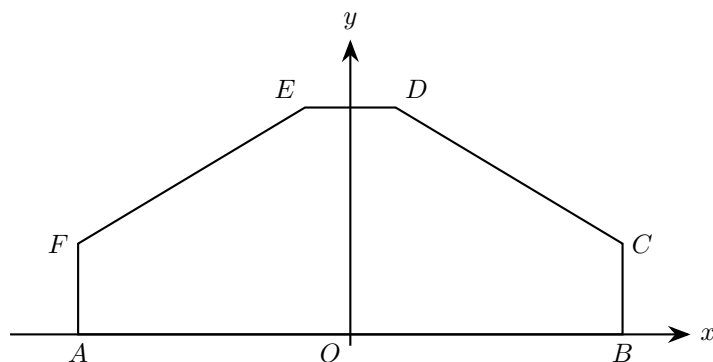
In part (a) you may use, without proof, any of the centre of mass formulae given in the formulae booklet.

- (a) Show that the distance of the centre of mass of triangle ABG from AB is $\frac{2a}{3}$. [3]

- (b) Find the coordinates of the centre of mass of the lamina. [4]

The lamina is freely suspended from F and hangs in equilibrium with FC at an angle θ° to the downward vertical.

- (c) Find the value of θ . [4]



10. A uniform lamina $ABCDEF$ is in the shape of the hexagon shown in the diagram. In a coordinate system with origin O , the vertices have coordinates $A(-120, 0)$, $B(120, 0)$, $C(120, 40)$, $D(20, 100)$, $E(-20, 100)$ and $F(-120, 40)$. The units of the axes are centimetres.

The centre of mass of the lamina lies at $(\bar{x}, \bar{y})$.

- (a) Use symmetry to write down $\bar{x}$ and show that $\bar{y} = 40$. [5]

The lamina is placed horizontally so that it rests on three supports, whose points of contact are at D , E and G , where G lies on the edge AB and has coordinates $(g, 0)$.

- (b) Determine the range of values of g for the lamina to rest in equilibrium. [3]

It is now given that $g = 8$, and that the lamina has weight 100 N .

- (c) Determine the forces exerted on the lamina by each of the supports at D , E and G . [4]